



ITS Network Security Study Sheet

CIA

- File Checksum checks for data integrity

Client Server

- Windows Server Update Services
 - Enables network admins (from the server) to deploy the latest Microsoft product (clients) updates.
 - Demonstrates a Client Server Relationship

Policies

- **Remote Access:** Cyber security policies governing network and computer access in the office, to remote (off-site) users wanting to connect to the network.
- **Access Control:** Policies that specify how access is managed and who may access information under what circumstances.
- **Incident Response:** A detailed document specifying how an organization prepares for and deals with cybersecurity incidents before and after they have taken place.
- **Acceptable Use Policy:** A Policy that outlines rules set by a network or service owner, governing the acceptable ways users can use the system.

- **Audit Policies:** logs events on the computer.



- Audit Logon Events: Logs each time the computer validates account credentials.
- Audit System Events: Logs each time someone reboots the workstation.

Backups

- Backups are best to be stored in an offsite location.
- **Full**
 - A full backup is a complete copy of a business or organization's data assets in their entirety. This process requires all files to be backed up into a single version.
 - FASTEST to restore
- **Incremental**
 - A backup that only copies data that has been changed or created since the last backup activity was conducted.
 - Longest to restore
- **Differential**
 - A differential backup collects data that has changed or been created since the last full (normal) or incremental backup, but it does not clear the archive bit on the file.
 - Medium in terms of speed to restore
- **Backup Steps:**
 - Backup the entire system

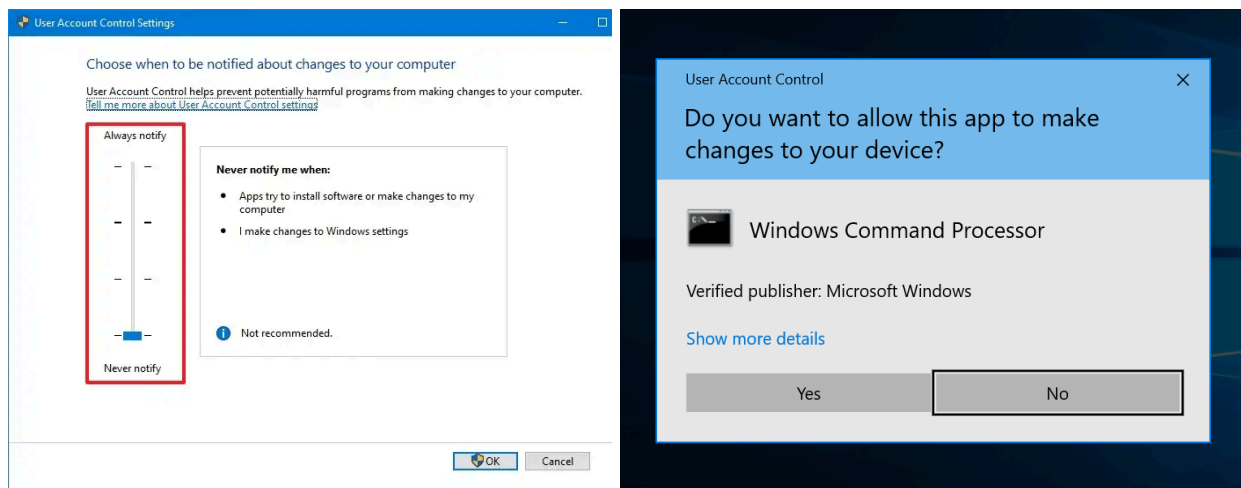
- Reformat the disk
- Reinstall the OS and applications from the original media
- Restore the user data from a backup image
- Update Everything including the OS and applications

NTFS Permissions

Permission	Meaning for Folders	Meaning for Files
Read	Permits viewing and listing of files and subfolders	Permits viewing or accessing of the file's contents
Write	Permits adding of files and subfolders	Permits writing to a file
Read & Execute	Permits viewing and listing of files and subfolders as well as executing of files; inherited by files and folders	Permits viewing and accessing of the file's contents as well as executing the file
List Folder Contents	Permits viewing and listing of files and subfolders as well as executing of files; inherited by folders only	N/A
Modify	Permits reading and writing of files and subfolders; allows deletion of the folder	Permits reading and writing of the file; allows deletion of the file
Full Control	Permits reading, writing, changing, and deleting of files and subfolders	Permits reading, writing, changing, and deleting of the file

User Account Control (UAC)

- UAC reduces the risk of malware by limiting the ability of malicious code to execute with administrator privileges.
- Reduces permission to that of a standard user unless higher permissions are necessary
- Notifies you when additional permission are required and asks whether you want to continue
- UAC can be disabled.
- (see Images below)



WIFI / Wireless Internet

- WPA2:

- WPA2 Enterprise: Strongest Form of Encryption that's typically used in a business setting
 - RADIUS Server (for AAA – authentication, authorization and accounting) for users needs to be in place for WPA2 Enterprise.
 - TACACS+ is another authentication system but specifically for admins.
- WPA2 PSK: Used in a home setting
- Uses AES encryption, which typically uses a strong symmetric key algorithm. (Can use TKIP as well if the user chooses to.)
- **WPA:**
 - Uses TKIP Encryption
- **WEP**
 - The weakest form of encryption
 - Uses a 40-bit, preshared encryption key. (Not Secure)
- Users of Wifi are generally exposed to eavesdropping and rogue access points.

NAT: Network Address Translation

- Implemented initially to preserve IP addresses
- Maps internal IP address to external IP addresses
- Located within a router.

DMZ / Perimeter Network

- A DMZ network (demilitarized zone network) is a perimeter network that protects an organization's internal LAN from untrusted traffic by logically separating the device from the internal network.

VPN: Virtual Private Network

- Ideally uses encapsulation and encryption to send data.
- Tunneling is the encapsulation of data as it moves from one place to another.
- Information goes through a VPN server first before going into the internet, which obscures (hides) the sender's identity (IP Address).
- **IPSEC:**
 - IPsec requires DOES NOT network applications to be IPsec aware
 - IPsec encrypts data
 - IPsec adds overhead for all network communications for which it is used

VLAN:

- A LOGICAL overlay network that groups together a subset of devices that share a physical LAN.

Firewall

- Stateless: This enables the firewall to perform basic filtering of inbound and outbound connections.
- Stateful: Stateful firewalls monitor all aspects of the traffic streams, their characteristics and communication channels.

Web Browsing Security

- Cookies: A benign (harmless) form of spyware that:
 - Tracks your web browsing habits
 - Enables storage of Website passwords
- Incognito: Hides Web Activity on a shared computer.
- Unsecured website uses HTTPS
 - Secure because it uses SSL or TLS certificates
 - SSL / TLS is a certificate that must be purchased from a trusted Certificate Authority.
- Block untrusted pop-ups.
- Pop-ups can display realistic OS or Application Errors (trojans)